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Serious game the Emotion Detectives helps to improve social–emotional skills of children with neurodevelopmental disorders

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Abstract

Children with neurodevelopmental disorders often have social–emotional difficulties. Serious games can be used to support their social–emotional learning. This study investigated if 6–10-year-old children with autism spectrum condition, attention deficit hyperactivity disorder and developmental language disorder would improve their emotion discrimination skills by playing a serious game. Intervention ($n=30$) and control groups ($n=20$), comprising children with neurodevelopmental disorders, and a group of typically developing (TD) age peers ($n=106$), completed six emotion discrimination tasks. ABA research design was applied in this study. Prior to the game intervention, children with neurodevelopmental disorders were significantly delayed in relation to their TD age peers in all tasks. After playing the game for two months, the children in the intervention group had significantly improved their emotion discrimination skills in four tasks, whereas the control group had significantly advanced in only one task without practice. Intervention gains were maintained at the one-month follow-up. Parents reported that the children in the intervention group somewhat improved their emotion recognition and behavioural skills in daily life in comparison to the control group. Implications for practice are discussed.

KEYWORDS

emotion discrimination, emotion recognition, game-based learning, serious games, social–emotional learning

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Practitioner notes

What is already known about this topic

- Children with neurodevelopmental disorders often exhibit various social–emotional symptoms.
- Within the field of social–emotional learning, many serious games have a narrow focus on specific skills, and they often only provide possibilities for observing target behaviours, not actual skill training.
- There are mixed results on the efficacy of game-based interventions targeted at improving social–emotional skills.

What this paper adds

- The Emotion Detectives game used in this study offers a well-designed learning environment that provides adaptive learning and feedback systems.
- The participating children's social–emotional skills were assessed with versatile methods and via triangulation.
- Intervention with the Emotion Detectives game helped to decrease some emotion and behavioural symptoms of the players.

Implications for practice

- This study provides information about the efficacy of a web-based game that is freely available for supporting children's social–emotional learning.
- Improvements in emotion discrimination and recognition, as well as behavioural skills, can be gained by playing serious games.
- For optimal intervention outcomes, multimodal training should be utilized in serious games targeting social–emotional learning.

INTRODUCTION

Neurodevelopmental disorders are conditions with onset in the developmental period which negatively affect child's brain functions and cause difficulties in cognitive and linguistic development. Autism spectrum condition, attention deficit hyperactivity disorder (ADHD) and developmental language disorder (DLD) are common conditions among neurodevelopmental disorders. The core symptoms of autism are challenges in communication and social interaction, language difficulties and stereotypical behaviour (Baron-Cohen et al., 2009). The central difficulties in children with ADHD lie in inattention, hyperactivity and impulsivity (Evans et al., 2018). Children with DLD, on the other hand, have language deficits which are not explainable by non-verbal abilities (Bishop et al., 2017). Children with neurodevelopmental disorders often have associated social–emotional and behavioural difficulties, and it is particularly common for these problems to co-occur with communication disorders (Sim et al., 2013; Yew & O'Kearney, 2013). There are many conventional methods to teach social–emotional skills (Bierman & Motamedi, 2015; Westby & Robinson, 2014), but the advantages of using digital technology are multiple. Serious games are a good example of motivational learning, as they provide feedback and endless possibilities for repetition. Additionally, serious games offer independence from time and place, cost-effectiveness and the possibility for multi-device use.

Recently, educational games (Vasalou et al., 2021) and, for example, augmented reality (Aydoğdu, 2022) have been successfully used to support young children's learning of

academic skills. Additionally, learning efficiency can be increased by adaptive digital technologies used in games (eg, Vasalou et al., 2021). Teens, in a Finnish study, for example, especially enjoyed games that contained problem-solving tasks (Kahila et al., 2020). Kahila et al. (2020) recommended that studying learning experiences does not suffice; one also needs to study actual learning outcomes or learning processes by using control groups or by using pre- and post-assessment designs for measuring learning. Peirce (2013) noted that there is a paucity of evidence on the effectiveness of children's serious games.

Many serious games are often developed for typically developing (TD) children, and in many cases, they can also be useful for children with developmental disorders. However, children with neurodevelopmental disorders often have, for example, language delays or short-term memory problems and therefore need targeted games suitable for them. There is still insufficient and mixed evidence on the effectiveness of serious games designed to improve children's social–emotional skills (Grossard et al., 2017; Kokol et al., 2020; Ramdoss et al., 2012; Saleme et al., 2020), and most of them target to improve the emotion recognition skills of children on the autism spectrum.

Supporting social–emotional skills using serious games

Social–emotional skills encompass a complex set of skills, such as recognizing, discriminating and labelling emotions, emotion regulation, large enough emotion vocabulary, and the ability to form and maintain friendships (Weissberg et al., 2015). Typically, children gradually learn social–emotional skills in social contexts from very early on, with language and Theory of Mind (Korkmaz, 2011) skills developing simultaneously. Poor social–emotional skills often result in, for example, problems in forming and maintaining peer relationships, compromised educational attainments, poor self-esteem, and behavioural problems, which may even persist into adulthood (Cohen & Mendez, 2009; Waddington et al., 2018).

Children with neurodevelopmental disorders are commonly known to have a wide range of social–emotional difficulties. Together with research concentrating on using other interventions, research on enhancing social–emotional skills through serious game-based interventions has thus far mostly focused on children on the autism spectrum (eg, Grossard et al., 2017; Ramdoss et al., 2012). However, children with ADHD and DLD have increasingly been shown to demonstrate similar difficulties to those with autism (Botting & Conti-Ramsden, 2008; Charman et al., 2015; Demopoulos et al., 2013; Taylor et al., 2015; Waddington et al., 2018).

Studies on the effectiveness of interventions focusing on social–emotional skills training in children with, for example, DLD are still sparse (Arts et al., 2022), and information about intervention outcomes using serious games with these children is even more so. However, although not all interventions have been found effective, in children with autism evidence exists on positive effects of social–emotional skills training carried out either with traditional methods or with digital technologies. In the review of Kokol et al. (2020), some promising results, on, for example, emotion recognition interventions using serious games were identified for children with autism. These results were achieved with games such as *EmotiPlay* (Fridenson-Hayo et al., 2017), which focuses on improving facial, vocal and bodily emotion recognition skills, and *JeStiMulE* (Serret et al., 2014), which targets training on facial emotion recognition skills. Positive effects were also found in a randomized controlled trial in which game-based intervention improved emotion recognition skills among children with autism (Rice et al., 2015). Regarding children with ADHD, research on the effectiveness of serious games has mostly explored the improvement of cognitive skills, such as time management and attention (Kokol et al., 2020). However, there is emerging research on also improving, for example, social skills in children with ADHD. In their review, Doulou and Skianis (2023)

identified some serious games to have obtained significant intervention outcomes on the social skills of children with ADHD. As of now, there seems to be only a few studies which report the use of serious games in social–emotional training in children with ADHD as well as DLD alone although their difficulties in social–emotional skills have been recognized.

Most studies often focus only on one modality of emotion recognition, for example, on only recognizing emotions from facial expressions, although some have included vocal or bodily recognition as well. The present study encompasses the assessment and intervention of discriminating and recognizing emotions not only from facial expressions but also from the voice and bodily postures as well as supporting the expansion of children's emotional vocabulary and improving positive problem-solving skills in complex social contexts. Furthermore, as children with neurodevelopmental disorders form a heterogenic group, as opposed to many previous studies, the present study included participants from three, often overlapping, diagnostic groups to yield results that may be more generalisable and reflect the reality clinicians and teachers face in their daily practice. To our knowledge, when studying the effectiveness of a serious game, no other study included all three diagnoses covered in the present study and also encompassed the multiple domains of social–emotional learning.

Background and basic characteristics of the Emotion Detectives game

The Emotion Detectives (ED) game (Huttunen et al., 2015; technical implementation by Oodles Solutions Ltd and Finnish National Agency for Education as the producer of the game; the game is freely available online at: tunneetsivat3.oph.oodles.fi) was used in the present study to support the learning of social–emotional skills. It was designed for children aged 6–12 years old to help them learn emotion recognition and discrimination skills, as well as social situation problem-solving skills (Huttunen et al., 2015; Koivula et al., 2017). The ED game was constructed to particularly meet the needs of children with neurodevelopmental disorders, hearing impairments and corresponding challenges, and its contents are based on previous research evidence (Brownell et al., 2013; Curenton & Craig, 2011; Tanaka et al., 2013) as well as clinical experience of, for example, speech therapists among the game constructors. The game provides modelling and training in discriminating, recognizing and labelling facial expressions, tones of voice and matching them.

The theoretical grounds of the ED game are based on the social information–processing model of Crick and Dodge (1994). This model emphasizes a bottom-up approach to social information decoding by suggesting that people use their previous experiences and integrate emotional processing into cognitive behavioural models. By playing the ED game, the children are expected to use their pre-existing cognitive resources and previous social experiences to interpret and utilize the social–emotional cues provided by the game. The pedagogical foundation of the game lies in the expectance that by making decisions in the game and getting feedback on the decisions made, children gain new experiences, providing them with possibilities to adopt new social behaviour models. The social information–processing model has also been used in developing a serious game by, for example, Hobbs and Yan (2008). The ED game used in the present study provides support for collaborative social skill learning, contains stable social rules and allows dynamic interaction with one of the game characters, a virtual assistant *Aksu*. To provide personalized feedback about a player's performance, *Aksu* gives positive spoken feedback (which also emerges as text) when the player succeeds in the tasks and encouragement when the player fails. This kind of use of a virtual assistant simulates children's typical language development process in which an adult scaffolds child's development by using positive emotion talk and mental vocabulary (Brownell et al., 2013; Curenton & Craig, 2011).

The ED game is designed to be adaptive (regarding the importance and role of this function, see, eg, Streicher & Smeddinck, 2016). When the player selects the adaptive game mode, the challenge level is increased by time pressure (eg, response required within five seconds). The adaptability of a learning game has been found to be beneficial and efficient for young children who are unable to self-regulate their learning (Debeer et al., 2021). Furthermore, a balance of challenges and skills can be considered important for creating psychological motivation.

The game is also constructed to be immersive, competitive and rewarding to provide enjoyability while also ensuring learning goals. The competitive nature of the game derives from the storyline according to which the player competes with other offices (see later the storyline). Additionally, the reward system of earning points supports the competitive nature of the game. The scoring system of the game consists of different prizes (eg, fame points) the player gets when they succeed at a certain point level in addition to being allowed to advance in the game. It is possible for the player to directly share the points earned on different tasks on social media.

According to the plot of the game, the player sets up an office with a virtual assistant, an imaginary creature called *Aksu* (Figure 1, *Aksu* is sitting on a flap folder cabinet). To provide gamified interaction and personalized feedback about a player's performance, the virtual assistant *Aksu* gives positive spoken feedback (which also emerges as text) when the player succeeds in the tasks. Using the characters and objects of the game, a player can also create their own cartoons with StoryTube, a cartoon generator programme, which is an extension to the ED game. The cartoons constructed can be saved and printed.

The goal of the game is to collect fame points by playing two types of exercises: “Office Tasks” and “Field Tasks”. The office tasks consist of 23 different minigames, such as puzzles, in which the player is required to match and label different emotions: happy, sad, angry, surprised, disgusted and neutral (Figure 2). The minigames focus on three domains of emotional learning: there are ten tasks of discriminating facial expressions (including matching of face and voice), six tasks of discriminating the emotional tone of voice and seven tasks of expanding emotion vocabulary. Vocabulary learning is further expanded by embedding metaphors (with their explanations) and other kinds of figurative language in the field tasks, instructions and other linguistic content throughout the game.

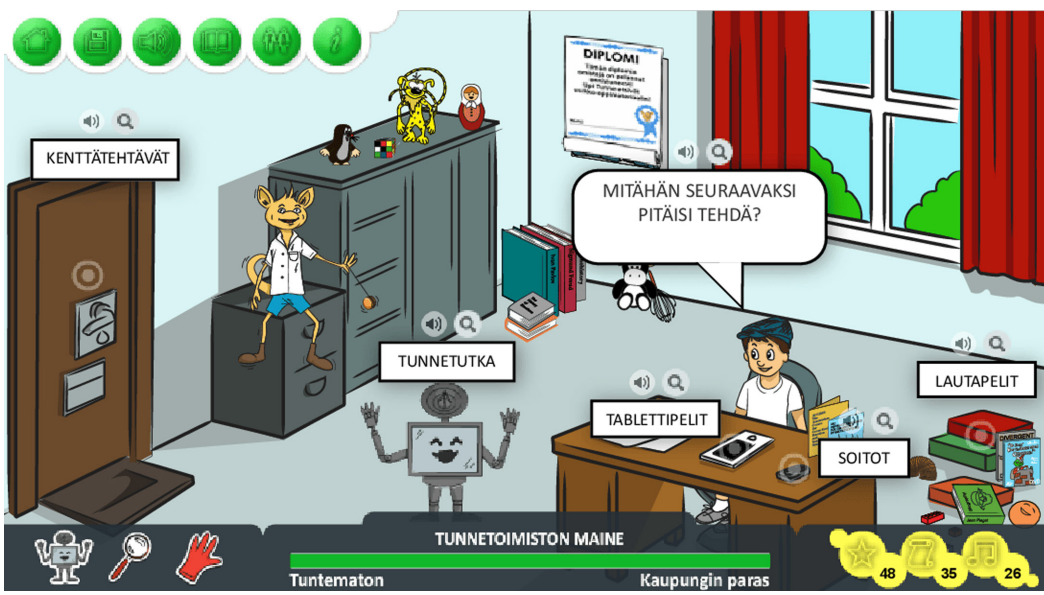

FIGURE 1 Office of the Emotion Detectives game.

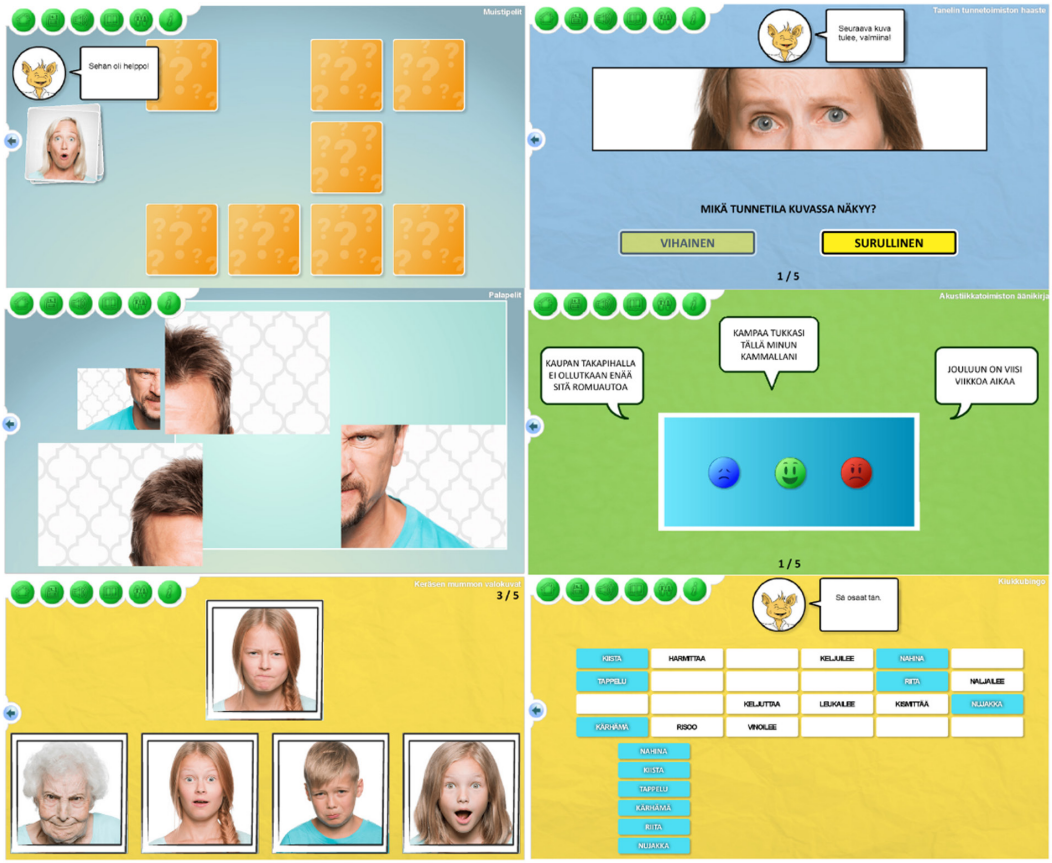

FIGURE 2 Examples of “Office Tasks” in the Emotion Detectives game.

It is impossible to advance in the ED game by playing only minigames. Therefore, the player is prompted to complete all the Field Tasks to progress further in the game. The Field Tasks include emotionally challenging social contexts and situations that many children often encounter in their daily lives (Figure 3). The social contexts include, for example, verbal misunderstanding, fears, loneliness, shyness, peer rejection, poor self-regulation, lying by a peer and so on. In written instructions, adults instructing and supporting children are provided with a list of ideas for possible conversation starters on how to address, elaborate and further link the child's own experiences to the specific social contexts they encounter during the game. The Field Tasks include 13 different social situations ordered by difficulty, in which the player needs to find out (1) which emotion is depicted by the characters in the game, (2) what the reason for the emotion is and (3) how the social problem can be resolved in a constructive way. All 13 Field Tasks comprised these three levels.

For the player to advance in the game, they need to have successfully played both the Office and the Field Tasks at a sufficient performance level. The points earned in three categories, facial expressions, emotional tone of voice and emotion vocabulary, are continuously updated on the screen. By gaining fame points, the player also earns tools, the Magnifying Glass and the Glove, which further help the player to advance in the game. When the player has earned enough fame points, the game is over with a rewarding ending story and congratulations. The player also gets a diploma that pops up on the wall of the office.

FIGURE 3 Example of "Field Tasks" in the Emotion Detectives game.

Aims

The first research question formulated in the present study is as follows: Do the children in the intervention group improve their emotion discrimination skills as a function of an intervention period during which they played a serious game—the ED game—and did they differ from the control group in the improvement of skills? The second research question explored whether the intervention group maintained its possible improvement in skills after the follow-up period. Finally, based on parental perceptions, with the third research question it was investigated whether the possible improvement of emotion discrimination skills in the intervention group would predict the improvement of daily life emotion recognition and behavioural skills.

METHOD

The study protocol was approved by the Ethical Committee of the Northern Ostrobothnia Hospital District, Finland, and it conforms to the Declaration of Helsinki as revised in 2000 (World Medical Association, 2001). The assent of the children and informed written consent from their parents were obtained before their participation in the study. Additionally, written consent was obtained from those children who were able to read the research notice, which had been specifically adjusted to their language level, and provide consent themselves. The data on children with neurodevelopmental disorders were gathered by direct testing and by enquiring parental perceptions, but the data on TD children were obtained only by direct testing.

The inclusion criteria used for the children with neurodevelopmental disorders to be recruited were as follows: (1) diagnosis of autism spectrum condition, ADHD or DLD; (2) difficulties in recognizing emotions from either the face, voice, or both, as reported by parents or speech and language therapists, occupational therapists, psychologists, early education teachers, or school teachers; (3) age between 6 and 10 years; (4) performance intelligence quotient over 85; (5) monolingual Finnish-speaking family; and (6) sufficient vision, hearing, motor and attention skills for test situations. Information needed for the children to fulfil these inclusion criteria was gathered from the parents. As exclusion criteria, the children were not allowed to have any psychiatric diagnoses or any ongoing interventions focusing on emotion recognition skills.

Participants

The children with neurodevelopmental disorders (called the clinical group) were recruited from different health-care settings, schools and parental organizations. They were allocated into the intervention ($n = 30$, five females, 14 children on the autism spectrum, seven with ADHD and nine with DLD) and the control ($n = 20$, four females, six children on the autism spectrum, 10 with ADHD and four with DLD) groups. Of the children on the autism spectrum, 17 out of 20 had been diagnosed with Asperger's syndrome. To get enough children to play the serious game, the participants for the intervention group were recruited first and the control group after that. Additionally, a group of TD age peers ($n = 106$, 20–22 children in each age group) was recruited from different day care centres and schools to serve as a neurotypical reference group for the self-constructed emotion discrimination tasks for which no published age norms were yet available. The mean age of both the children with neurodevelopmental disorders and the TD children was 8.02 years. In the clinical group, neither the age nor the male: female ratio significantly differed by diagnosis. However, the number of males was significantly higher than in the TD age peer group (Fisher's exact test, $p < 0.001$).

Of the children with neurodevelopmental disorders, 30 children had a single diagnosis representing some neurodevelopmental disorder and 20 children had one or two comorbid diagnoses belonging to this entity. Of the children with comorbid diagnoses, six had been diagnosed with both autism and ADHD, seven had ADHD and DLD, three had autism and DLD and four had all three diagnoses. In the case of comorbid diagnoses, the primary diagnosis was determined based on a child's medical records provided by parents and scientific and clinical literature. For example, if diagnoses of both autism and ADHD had been given to the same child, autism was determined to be the primary one as based on clinical routine in Finland. If a child had medication for ADHD and he or she also had DLD, ADHD was judged to be more severe and therefore the primary diagnosis.

In our earlier study (Löytömäki et al., 2020), we did not find any statistically significant differences in expressive vocabulary of these children included in the three diagnostic groups. The Boston Naming Test (Kaplan et al., 1983) was used in testing these children. It is established that vocabulary reflects a child's overall linguistic development.

Data collection and instruments

Data collection set of this study comprised six emotion discrimination tasks the children accomplished, and two questionnaires addressed to parents. All six emotion discrimination tasks and parental questionnaires were used both in the pre- and post-assessments for both intervention and control groups. At the follow-up assessment phase, only the intervention group was tested with the emotion discrimination tasks and only the parents of the intervention group were asked to fill out the questionnaires.

The children were tested individually at home, at school or in a clinical setting, and the assessment sessions were audio- and video-recorded for closer analyses. Polanczyk and colleagues (2015) recommended that it is desirable to collect data from different sources. In the present study, triangulation was used, as the data were collected both by direct testing and by collecting parental views on their child's skills and behaviour in daily life.

The term emotion discrimination is used in this study when referring to discerning emotions when alternative choices are given for answering, and the term emotion recognition is used when referring to emotion perception in everyday life. Overall, six tasks were used to explore the children's emotion discrimination skills. Five were specially constructed for this study, and one was a published test (the FEFA 2 test; the Frankfurt Test and Training of Facial Affect Recognition; Bölte & Poustka, 2003; Finnish version: Bölte et al., 2013). To

determine the age-appropriate level of performance, TD age peers ($n = 106$) also completed the same tasks, except for the FEFA 2 test, as it was already a published test.

To explore discrimination of emotional tone of voice, two tasks were constructed because no standardized tasks in the Finnish language were available. The first task focused on nonsense words, half embedded and half not embedded in a carrier sentence of neutral content ("Now I say…"), carrying an emotional tone of voice. The second task focused on sentence-level and had semantic meaning which matched the emotional tone of voice. In the emotional nonsense word task, the children listened to 18 items spoken by a female professional actor recorded in a professional recording studio. Three emotions—happy, sad and angry—were included in this task with each of the emotions emerging three times in a random order. The task containing meaningful sentences comprised six basic emotions and ashamed and neutral voices. In both tasks, the participants could answer either verbally after being given the response alternatives or by pointing to an emoji placed in front of them.

To assess the discrimination of facial expressions, two tasks were constructed: one containing eight static photographs and the other containing eight video clips. Both tasks included the so-called six basic emotions—joy, anger, sadness, fear, surprise and disgust—and additionally, ashamed and neutral facial expressions. The emotions were posed in a professional photographer's studio by two adults (professional actors) and two children. In both tasks, the test administrator provided the children with four verbal labels as response alternatives, reading them aloud. In addition to the two self-constructed tasks, the Faces submodule of the Finnish version of the computerized FEFA 2 test was used to explore the discrimination of facial expressions. The Faces submodule consists of 50 photographs depicting seven different emotions (six basic emotions and a neutral facial expression). The original German version of the test has been reported to have excellent psychometric properties (Bölte & Poustka, 2003).

Finally, a task in which facial expressions and emotional tone of voice had to be matched was constructed and presented. Multimodality has been found to improve discrimination responses (Gerdes et al., 2014). This task comprised 11 different items (including, again, the six basic emotions and ashamed and neutral expressions) and consisted of photographs and voice samples representing the same emotions that were included in the task in which the emotional tone of voice needed to be discriminated in sentences. Children heard a sentence with an emotional tone of voice and then pointed at a photograph depicting a facial expression with which they thought the sentence they heard matched.

The parents and caregivers of the children in the intervention and control groups were asked to provide information about the children's emotion recognition skills and behaviours at the pre- and post-assessment phases. Parents assessed the abilities of their children to recognize emotions from the face, voice and bodily expressions using a 10 cm visual analogue scale (VAS). Higher score on VAS represented better skills. The parents of the intervention group were also asked to evaluate how much they played the ED game together with their children as a percentage of the total play time.

The Strengths and Difficulties Questionnaire (SDQ; Goodman, 1997) was used to explore how parents perceived the behavioural problems and skills of their children. The SDQ is a 25-item Likert-type inquiry about 4–17-year-old children's and adolescents' psychosocial abilities and difficulties. The SDQ has also been validated for Finnish 4–9-year-olds (Borg et al., 2014) and 7–12-year-olds (Koskelainen et al., 2000), with its reliability being acceptable. The questionnaire has five subscales: emotional symptoms, behavioural problems, hyperactivity, peer relationship problems and prosocial behaviour, each comprising five items. The answers to the subscale questions are scored from 0 to 2, and a total difficulty score can be generated, with total scores ranging from 0 to 40. A higher total score means more severe problems: at or above 12 points means an abnormal result according to the Finnish cut-off points (the 90th percentiles, as Goodman, 1997 recommends), based on the reports of 2635 parents and 2242 teachers provided by Borg et al. (2014).

Research design and intervention protocol

Reversal protocol (ABA research design) was used in the present study. In exploring the effectiveness of the ED game, both the intervention and control groups participated in the pre-assessments. After the assessment, the children in the intervention group started their two-month intervention by playing the game, while the control group did not play. After the two-month period, both groups participated in the post-assessments. Additionally, to explore the maintenance of the possible progress of skills gained with the help of the intervention, the intervention group participated in a follow-up assessment one month after the post-assessment.

The children in the intervention group were requested to play the game at home, at school or in a clinical setting (eg, during speech therapy sessions). The participants were advised to play with an adult, such as a parent, speech and language or occupational therapist or a school aid. Each child was provided with individual credentials for logging in. The researchers moderated playtimes weekly, and the adults were provided with their own credentials to keep track of the children's playtime themselves. The children were requested to play for one hour, at a minimum, and for two hours, at a maximum, each week over eight weeks (8–16 hours in total). This targeted play time was based on research conducted by Baron-Cohen et al. (2009), who mentioned that two weekly hours of training was enough in their earlier study to improve the recognition of facial expressions in children with autism. In the present study, if the researchers (the first and the last authors of the present article) noticed that the minimum playtime had not been reached by Friday afternoon, the families were sent a reminding text message with a gentle prompt to urge the child to play more before the end of the week. If a maximum playtime of two hours was attained during any single week, the game would not allow the child to continue until the next week began.

The player can choose the difficulty level in the ED game: easy, normal, demanding or adaptive. To facilitate maximal learning, the participants in the present study were required to choose only the adaptive game mode. The adaptive mode allowed the child to progress in the game only after the player had reached a sufficient level of learning (eg, 80% of the Office Tasks accomplished at a sufficing level).

To make playing possible for illiterate children, almost all the instructions and texts in the speech bubbles of the ED game characters could also be listened to by clicking the loudspeaker icons on the screen. All text was spoken by the game's narrator's voice. Despite the possibility of listening, minor parts of the game still required reading ability. Therefore, especially the parents of illiterate children were advised to play together with their children. All vocal and photo materials were posed by children and adult actors and were recorded and edited in a professional sound-recording studio and a professional photographer's studio.

Statistical analyses

SPSS (IBM SPSS Statistics for Windows, Version 27.0. Armonk, NY: IBM Corp.) and SAS (version 9.4, SAS Institute Inc., Cary, NC, USA) were used for the statistical analyses. Between-group comparisons were performed using Fisher's exact test for categorical data and analysis of variance (ANOVA) or Student's *t*-test for continuous data. ANOVA was used if there were more than two groups to be compared. Bonferroni correction was used for post hoc correction in the ANOVA. Linear regression analysis was applied to determine if improvement in emotion discrimination skills would predict improvement in emotion recognition skills in everyday life, as assessed by parents with VAS, and in behavioural skills, as assessed by the SDQ total score. Linear regression analysis was performed only if the

correlation coefficient between emotion discrimination skills and emotion recognition skills in everyday life was >0.3 or <−0.3. Spearman's correlation coefficient was calculated to explore the associations between the continuous variables. Linear mixed-model ANOVA was used for repeatedly measured continuous data (eg, emotion discrimination tasks). Time, Group and Time × Group interaction were used as fixed effects when groups were compared and only Time when comparing change within Group. Participant was used as the random effect, and the variance component was chosen for the covariance structure. The Kenward-Roger method was used to calculate the degrees of freedom in the linear mixed model. The paired samples *t*-test was used for exploring the difference between the two time points for the control group. For testing the internal consistency of the tasks used, the KR-20 (for dichotomous variables) and Cronbach's alpha (for continuous variables) were used, and for testing the reliability of the measures, the Intra-class Correlation Coefficient (ICC) was used.

RESULTS

First, results on internal consistency and test–retest reliability are presented. The internal consistency results of the emotion discrimination tasks for the clinical group before the intervention were as follows: KR-20 was 0.52 for nonsense words, 0.54 for meaningful sentences, 0.36 for photographs, 0.39 for video clips and 0.57 for matching. The internal consistency of the three parent-assessed emotion recognition skills using VAS (intervention and control groups together, $n=49$) was good ($\alpha=0.73$), and the same was observed for SDQ ($\alpha=0.77$).

The test–retest reliability ICC (test repeatability) for the intervention group between post-assessment and follow-up assessments was 0.71 for nonsense words (95% confidence interval [CI], 0.48–0.86); 0.71 for meaningful sentences (95% CI, 0.46–0.85); 0.88 for the FEFA 2 test (95% CI, 0.77–0.94); 0.52 for photographs (95% CI, 0.19–0.74); 0.73 for video clips (95% CI, 0.49–0.86); and 0.69 for matching task (95% CI, 0.44–0.84). For the control group, between pre- and post-assessments, ICC was 0.40 for nonsense words (95% CI, −0.11 to 0.71); 0.71 for meaningful sentences (95% CI, 0.41–0.88); 0.75 for the FEFA 2 test (95% CI, 0.47–0.89); 0.58 for photographs (95% CI, 0.18–0.81); 0.18 for video clips (95% CI, −0.30 to 0.58) and 0.54 for matching task (95% CI, 0.15–0.78).

Emotion discrimination skills and their improvement

For answering the first research question, the results on the skill improvement of the children in the intervention group will be described next. It was first explored whether the children in the intervention and control groups were delayed in their emotion discrimination skills when they were compared with the results of TD age peers before the intervention period. ANOVA showed that TD children's skills were better than those in the intervention or control groups. There was a significant difference between the intervention ($n=30$) and TD age peer ($n=106$) groups in all six tasks, and the control group ($n=20$) was significantly delayed in two tasks compared to the TD children. Both the intervention and the control groups were delayed in discriminating tone of voice in nonsense words, $F(2)=7.43$, $p=0.001$ and in the FEFA 2 test, $F(2)=17.21$, $p<0.001$, but only the intervention group was delayed in discriminating tone of voice in meaningful sentences, $F(2)=5.75$, $p=0.004$, in discriminating facial expressions from photographs, $F(2)=16.49$, $p<0.001$, in discriminating facial expressions from video clips, $F(2)=9.96$, $p<0.001$ and in matching facial expressions with emotional tone of voice, $F(2)=9.49$, $p<0.001$. The results of the intervention, control and TD age peer groups can be found in Figure 4.

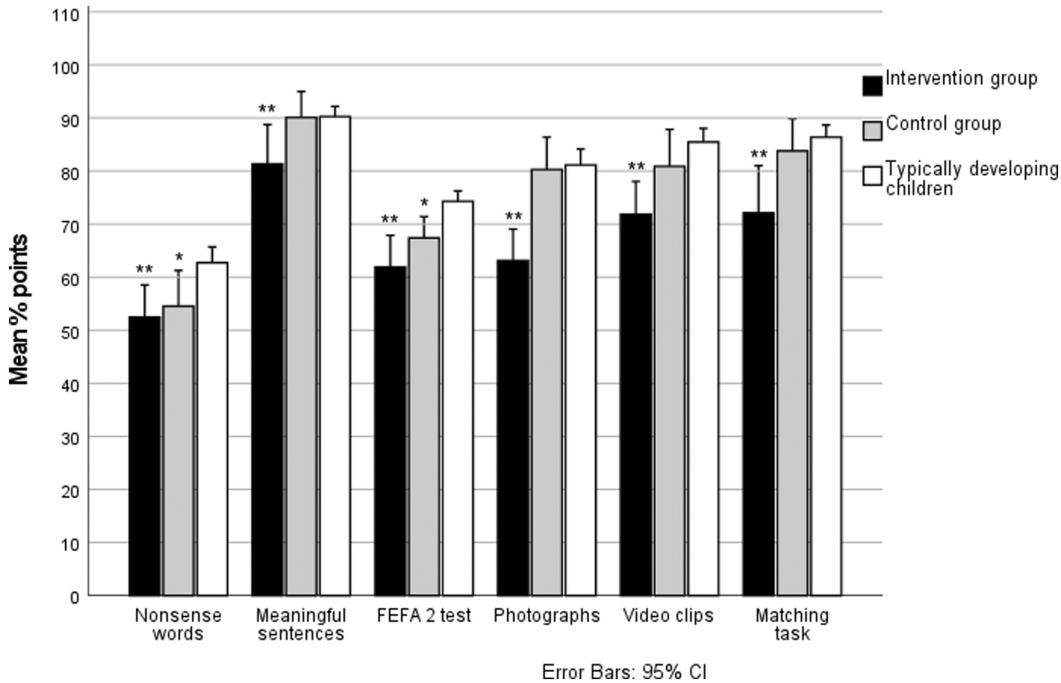

FIGURE 4 Emotion discrimination task results of the three groups at pre-assessment phase (means with 95% CIs). *Note:* **Significant differences between the intervention group and TD children; *Significant differences between the control group and TD children.

When comparing the children in the intervention group ($n = 30$) to the TD age peers, they were slightly, but statistically significantly delayed in all six emotion discrimination tasks. The largest difference, 18 percentage points, on average, was in the task of discriminating facial expressions from photographs ($t(132) = -5.56$, $p < 0.001$, $d = 1.17$) and the smallest, nine percentage points, on average, in the task of discriminating the emotional tone of voice in meaningful sentences ($t(33.2) = -2.31$, $p = 0.027$, $d = 0.68$). However, the children in the control group ($n = 20$) were significantly delayed compared to the TD age peers in only two tasks: in discriminating emotions in nonsense words ($t(123) = -2.41$, $p = 0.018$, $d = 0.59$) and in the FEFA 2 test ($t(123) = -3.04$, $p = 0.003$, $d = 0.74$).

Finally, when the intervention and the control groups were compared with each other, they were found to perform similarly at pre-assessment in all emotion discrimination tasks except in the task of discriminating emotions from photographs, in which the control group performed significantly better than the intervention group ($t(47) = -4.04$, $p < 0.001$, $d = 1.18$).

When looking at the effectiveness of the game-based intervention, first, total play time was examined. Children in the intervention group ($n = 30$) played the ED game for, on average, 456 minutes, that is, 7 hours 36 minutes (*SD* 112, range 243–657), which is a bit less than the targeted minimum of eight hours per child. However, the mean playtime per week was 58 minutes (*SD* 15), which was almost the targeted one hour. Weeks with no or very little (less than 15 minutes) playtime represented only 12% (29 weeks out of 240) of the total number of weeks played by all the children. Overall, 14 children out of 30 had a mean of 2.1 weeks (*SD* 1.07, range 1–4), during which they did not play at all or played less than 15 minutes per week. For the first playthrough, which all children completed, there was an average of 103 (*SD* 36, range 47–192) completed tasks. Some children completed more than one playthrough. When enquired, the parents of 22 out of 30 children reported having played together with their children. They had done so, on average, in 59% (*SD* 38, range 5–100) of their children's total playtime.

TABLE 1 Playthroughs by task type of the children in the intervention group ($n = 30$).

	Mean	*SD*	Min	Max
Single Office Tasks, number of playthroughs				
Discrimination of facial expressions	32	18	9	96
Discrimination of tone of voice	21	17	6	93
Emotional vocabulary	19	15	3	81
Single levels of Field Tasks, number of playthroughs	80	32	25	142

Note: Office Tasks = 23 different minigames of emotion discrimination (10 for discriminating facial expressions, six for tone of voice and seven for practising emotion vocabulary); Field Tasks = 13 tasks each having three levels in which the player needs to solve complex social–emotional situations.

The amount of practice realized was specified by the task types completed (the Office and Field Tasks). The number of playthroughs the children in the intervention group successfully completed in each task type and, additionally, the subcategories of the Office Tasks can be found in Table 1. The number of successfully completed task types was not associated with the emotion discrimination skills tested at pre-assessment phase.

Next, it was explored whether the intervention group had improved in emotion discrimination skills after the intervention and whether there were any differences in the level of improvement compared to the control group. According to the mixed-model ANOVA, significant main effects of Time were found in four emotion discrimination tasks: nonsense words, $F(2, 57.2) = 30.13$, $p < 0.001$; meaningful sentences, $F(2, 36.3) = 3.77$, $p = 0.032$; FEFA 2 test, $F(2, 41.4) = 6.17$, $p = 0.005$; and photographs, $F(2, 56.9) = 3.71$, $p = 0.031$. Significant main effects of Group were found in two emotion discrimination tasks: photographs, $F(1, 47) = 14.61$, $p = 0.001$, and the matching task $F(1, 38.8) = 4.61$, $p = 0.038$, with the intervention group outperforming the control group. Time × Group interactions were found in two emotion discrimination tasks; compared to the control group, children in the intervention group significantly improved in their skills in two tasks: in discriminating the emotional tone of voice in meaningful sentences $F(1, 47.9) = 6.96$, $p = 0.011$ (mean pre-assessment result for this task in percentage points was 82; for post-assessment, it was 90) and in discriminating facial expressions in the FEFA 2 test $F(1, 42.4) = 4.26$, $p = 0.045$ (pre-assessment mean was 62; post-assessment mean was 68). Total playtime did not significantly correlate with the improvement in emotion discrimination skills measured with the six discrimination tasks.

Children in the intervention group with better and poorer pre-assessment results in emotion discrimination skills were able to advance similarly in the game. The number of times the children successfully completed the tasks of discriminating emotional tone of voice in the game significantly correlated with the mean improvement in discriminating emotions from nonsense words (Spearman's correlation coefficient [$r_s = 0.37$, $p = 0.048$]). No other correlations were detected between the amount of practice and the improvement in the emotion discrimination skills of the children in the intervention group.

Concerning the children in the control group, in the post-assessment, they had significantly improved in only one task: in discriminating emotions from nonsense words, $t(19) = -3.04$, $p = 0.007$. The differences in the improvement of children's skills in the intervention and control groups between the pre- and post-assessments can be found in Figure 5.

Maintaining of skills after intervention period

The results of the second research question in finding out whether the children in the intervention group maintained their newly learned emotion discrimination skills are described next. Mixed-model ANOVA found no statistical significance between the post-assessment

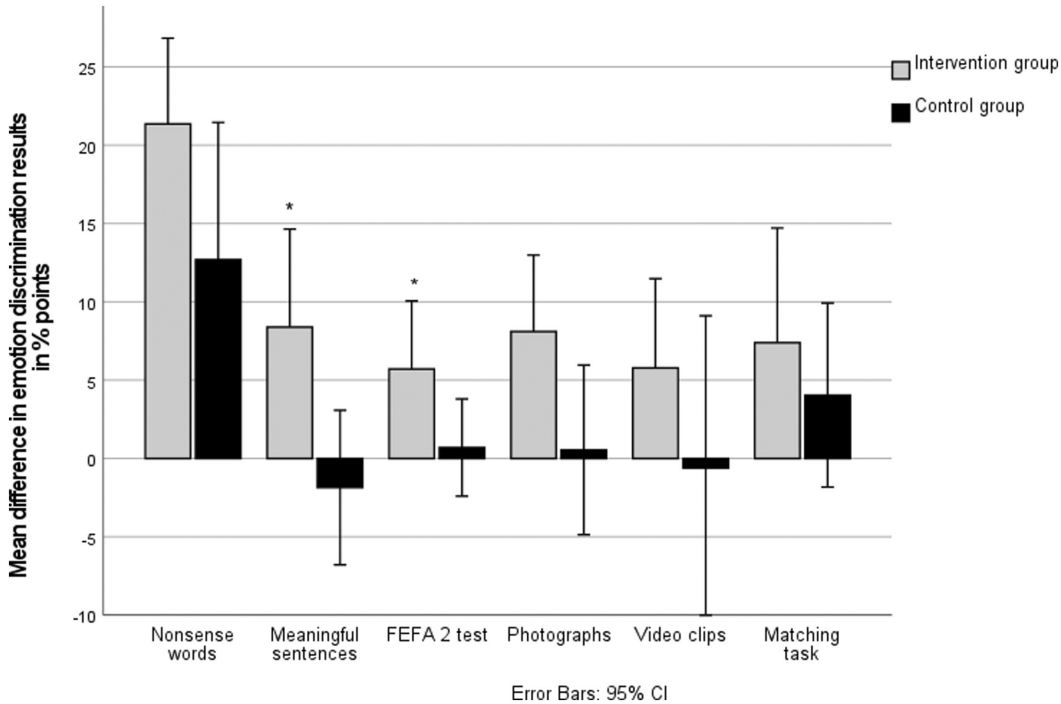

FIGURE 5 Change in emotion discrimination task results expressed as percentage points compared to the pre-assessment phase. *Note:* *Tasks in which the improvement of the skills of the intervention group significantly outperformed the improvement of the control group.

and follow-up assessment results in the intervention group, indicating that the children maintained their performance levels gained through the intervention.

Improvement of emotion discrimination predicting emotion recognition and behavioural skills in daily life

The third research question of the present study was to determine whether the parent-assessed daily life emotion recognition and behavioural skills of the children in the intervention group improved after the intervention. The improvement was also looked at in relation to a possible change in the tested emotion discrimination skills as a function of the intervention with the ED game. At the pre-assessment, as assessed by parents with VAS, the mean scores of the children were 58.21 for recognizing emotions from facial expressions, 58.91 for recognizing tone of voice and 55.00 for recognizing bodily postures (on a scale of 0–100; higher score meaning better skills). As for the SDQ scores, the mean total score was 17.37 (*SD* 5.6), and 27 out of 30 children scored at or above the 90th percentile (an abnormal result when compared to the Finnish norms published by Borg et al. [2014]).

After the intervention with the ED game in the intervention group, the mean parent-reported VAS assessments were 57.21 (*SD* 24) for recognizing emotions from facial expressions, 68.10 (*SD* 25) for recognizing tone of voice and 62.52 (*SD* 24) for recognizing bodily postures. The improvement, however, was not significant for any of the VAS assessments. For the SDQ total score, the mean post-assessment result was 15.10 (*SD* 5.9), and the improvement as opposed to the pre-assessment was significant, $t(29) = 4.21$, $p < 0.001$, $d = 1.17$.

According to the linear regression analysis, no significant models were found for the improvement of daily life emotion recognition skills. For the SDQ total score, the best predictive model was found for the improvement of skills in discriminating tone of voice from meaningful sentences, which was related to the improvement of overall behavioural skills as measured with the SDQ total score. The result was significant $F(1, 24) = 4.38$, $p = 0.046$, and this model predicted 14.4% of the variation. Sex and age were not significant predictors in any of the models.

DISCUSSION

For answering the first research question, it was found out that with the help of the ED game, the children in the intervention group, who, compared with the TD age peers, were significantly delayed in their emotion discrimination skills, significantly improved their skills in four out of six tasks. Compared with the children in the control group who did not play the game, the improvement of the children in the intervention group was significantly greater in two tasks. Those serving as controls did not practically advance their emotion discrimination skills; their improvement was significant in only one task. Improvements in social–emotional skills have also been discovered in previous studies on gamified interventions. For example, Craig et al. (2016) used game-based social–emotional training and found greater improvement in the skills of the treatment group compared to the control group. However, of their overall participants, only about one-fifth had emotional and behavioural problems. In two systematic reviews (see Grynszpan et al., 2014; Ramdoss et al., 2012), mixed results regarding the effectiveness of game-based social–emotional intervention were found. While many of the studies in these reviews could not confirm the efficacy of the games in social–emotional learning, a slight majority of them were still able to show improvement in social–emotional skills among individuals with autism.

The answer to the second research question was that after the one-month follow up, children had maintained their newly acquired emotion discrimination skills. This result suggests that emotion discrimination skills can be acquired with consistent training using a serious game and also maintained at least for a rather short period of time.

Finally, for answering the third research question, improvements in behavioural skills were also found in the present study. The improvement in the parent-assessed SDQ total score on behavioural problems in the intervention group was explained by the improvement in discriminating emotions from meaningful sentences. Previous research has shown that social–emotional difficulties and behavioural problems often co-occur (eg, Mackie & Law, 2010; Thomas et al., 2018), and the result of SDQ in the present study suggests that there may indeed be a link between social–emotional and behavioural difficulties. Therefore, it is an exciting thought that training skills focusing only on emotion discrimination and recognition may be further beneficial for reducing behavioural symptoms. However, the results of the present study remain clinically rather small, as they are based on small, although statistically significant, improvements.

Limitations

This study had some limitations. First, with the help of playing the ED game, the intervention group improved their emotion discrimination performance, but even though the improvement was statistically significant between pre- and post-assessments, it was still, in practice, relatively small. Secondly, the follow-up assessment was conducted as soon as one month after the post-assessment, which is a short time span considering the desirable long-term outcomes of the intervention. On the other hand, especially in children, there is a risk of maturation and other intervening factors when using a longer follow-up timeframe.

Thirdly, the biggest limitation of this study was the relatively small number of participants, although not unusual in this field of research. Especially the number of children with DLD in the intervention and control groups was only nine and four, respectively. That reduces the power of the present study and generalizability of its results. Finally, the intervention ($n = 30$) and control groups ($n = 20$) had an uneven number of participants, and the participants were not randomly assigned to the groups. This was for practical reasons, as, when recruiting, children who fulfilled the inclusion criteria were difficult to find. To secure enough children for the intervention group, they were prioritized when recruiting.

CONCLUSION

The present study reported the results of training emotion discrimination and recognition skills in children with neurodevelopmental disorders by using a serious game. Children in the intervention group improved their emotion discrimination skills during the two-month intervention period playing the ED game, with their progress being better than that in the control group. Participants of the intervention group had also maintained their skills when tested one month after the intervention. They also progressed in their behavioural and in daily life emotion recognition skills. Therefore, serious games can be considered useful tools in the training of social–emotional skills focusing on multimodal emotion discrimination. However, further research is needed to confirm intervention outcomes in larger child groups.

The present study showed that the children with autism spectrum condition, ADHD and DLD benefit from similar gamified intervention methods and future studies should therefore include all these groups of children when studying the effects of intervention on social–emotional skills. However, further research is needed to confirm intervention outcomes in more severe conditions and in larger child groups. It is also important to find out the optimal time practised with a serious game that is needed for achieving good learning outcomes. That information can be used in game intervention recommendations. Furthermore, it needs to be explored whether skills learned through gamified intervention are maintained over a longer than the one-month follow-up period we used.

Generalization of skills, one of the major challenges of any intervention, is necessary to be measured in intervention studies on game-based learning. We used two measures, questions addressed to adults on children's emotion recognition abilities in daily life and SDQ, but it would be beneficial to use even a broader battery. When examining real-life experiences and behavioural outcomes, the self-assessment of older children also needs to be included. Children may have different aims and appreciate different outcomes than adults and their self-perceptions on social–emotional skills may also differ from the views of their parents, teachers and therapists. Moreover, children's viewpoints on game contents and design can help game developers to construct more multifaceted, attractive and effective tools for improving children's skills.

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CONFLICT OF INTEREST STATEMENT

The authors report no conflict of interest.

DATA AVAILABILITY STATEMENT

Due to the identifiable and confidential nature of this data, the data are not available to other researchers.

ETHICS STATEMENT

The study protocol was approved by the Ethical Committee of the Northern Ostrobothnia Hospital District, Finland, and it conforms to the revised Declaration of Helsinki (2000). Written informed consent was obtained from parents and from those children who were able to read the research notice and sign the consent form.

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